

strategy card, which the other players had laughed at when I first sat down, was doing pretty well. I was holding my own.

What was the strategy that the article had recommended?

It was what is now known as the basic strategy in blackjack.⁶ After about 20 minutes, I got a hand with an ace and seven, a soft 18, and the dealer had a nine showing. The strategy said hit. The other players groaned when I did, thinking I was a fool. The next card was a four, so I had 12. The other players thought I had gotten just what I deserved. I then got a string of aces and twos, which gave me six cards with a total of 16. The strategy said hit again, and I got a five, which gave me a seven card 21. Now the other players really got excited. They thought that the strategy card was magical rather than stupid. I played for a while longer, and eventually ended up losing 8 of my 10 dollars. When I went back to UCLA, I reread the article, and I realized that there was a potential for beating the game by keeping track of the cards played.

I guess the article assumed no knowledge of the cards played with all outcomes equally likely on every deal.

Exactly, the article assumed a full-deck strategy all the time. On average, that is the right assumption, but if you have more information, like what cards have been played, then you can improve on those odds.

I was convinced that it was very likely that there would be significant swings in the edge back and forth. The question was how to identify those swings. I wrote the authors of the blackjack article, and they sent me all their lab manuals with the calculations. I spent about five or six weeks mastering all the details of what they had done. My plan was to repeat their calculations with some cards removed from the deck. For example, take out the twos, take out the threes, and so on. I was using a desk calculator and making very slow progress. I decided to estimate how long my calculations would take. I found was that if I did everything I wanted to do, it would take me several thousand years. By that time, I had taken a job at MIT. At the time, MIT had an IBM 704 computer, which was then the best available commercial computer.

What year was this?